ELEC3705 Fundamentals of Quantum Engineering

Laboratory 5 : Bouncing particles in a potential well

Abstract

We will simulate the time-evolution of a single electron confined in a small, 1-dimensional box by utilising a general solution to the Schrödinger equation.

Introduction

The Schrödinger equation in 1 spatial dimension is defined as

$$i\hbar\frac{\partial}{\partial t}\ (x,t) = \left[\frac{-\hbar^2}{2m_0}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\ (x,t) \tag{1}$$

where

$$\hbar = 1.05 \times 10^{-34}\,\mathrm{J\,s}$$
$$m_0 = 9.11 \times 10^{-31}\,\mathrm{kg}$$

(x,t) is the electron wavefunction and $V(x,t)$ is an applied electric potential which we will assume is constant. By solving the Schrödinger equation assuming the spatial and time components are separable we arrive at the state vector

$$(x,t) = \varphi(x)\exp(-i\omega t) \tag{2}$$

where $\exp(-i\omega t)$ constitutes a global phase factor and $\varphi(x)$ is a stationary state of our system defined by

$$\left[\frac{\partial^2}{\partial x^2} + \frac{2m_0}{\hbar^2}\Big(E - V(x)\Big)\right]\varphi(x) = 0. \tag{3}$$

Here, the electron described by $\varphi(x)$ has energy E. For an electron in a potential well with width a, we define $V(x)$ as follows

$$V(x) = \begin{cases} 0, & 0 < x < a \\ \infty, & \text{otherwise} \end{cases} \tag{4}$$

We can then find the general solution to equation 3 to be

$$\varphi(x) = \sqrt{\frac{2}{a}}\sin(k_n x) \tag{5}$$

with corresponding energy and wavenumber

$$E_n = \frac{k_n^2 \hbar^2}{2m_0}$$
$$k_n = \frac{n\pi}{a}$$

As we are choosing our computational basis as the eigenstates of our system, our Hamiltonian is a diagonal matrix and hence the unitary time evolution operator is also diagonal.

$$U(t,0) = \begin{pmatrix} e^{-iE_1 t/\hbar} & 0 & \cdots & 0 \\ 0 & e^{-iE_2 t/\hbar} & \ddots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & e^{-iE_N t/\hbar} \end{pmatrix} \tag{6}$$

Dispersion and group velocity

The dispersion relation, $\omega(k)$, describes the dependence of a wave's angular momentum on its wavenumber. In the case presented above, the energy of each wavenumber was $E(k) = E_n = \frac{k_n^2 \hbar^2}{2m_0}$, which we can relate to angular momentum through the Planck-Einstein relation ($E = \hbar\omega$), giving

$$\omega_n = \frac{k_n^2 \hbar}{2m_0} \tag{7}$$

The significance of this relation can be seen when looking at the phase velocity of a wavepacket composed of multiple frequencies or wavenumbers.

$$v_p = \frac{\omega(k)}{k} \tag{8}$$

A dispersionless medium is one in which the phase velocity is the same for any component frequency, i.e. each component travels through space at the same speed, and so the wavepacket won't spread out.

Question 1

Would a wavepacket created in the 1-dimensional box described above disperse?

Task

Given the following initial states (ensure you **normalise** $|\psi\rangle$!), simulate the time evolution of an electron in a potential well with infinite barriers and a width of 1 µm. Your system will be truncated to $N = 10$ levels.

a) $|\psi(t=0)\rangle = |\varphi_1\rangle + |\varphi_2\rangle$

b) $|\psi(t=0)\rangle = |\varphi_{n-1}\rangle + |\varphi_n\rangle$

c) $|\psi(t=0)\rangle = |\varphi_1\rangle + |\varphi_3\rangle + \cdots |\varphi_{2k+1}\rangle$ (only odd terms)

d) $|\psi(t=0)\rangle = |\varphi_1\rangle + |\varphi_2\rangle + \cdots |\varphi_n\rangle$

Extended Task

Try to construct a Gaussian wave packet from the eigenstates of your potential well. A Gaussian wave packet takes the form

$$|\psi\rangle = \sum_n \exp\left(-\frac{1}{2\sigma^2}(n-n_0)^2\right)|\varphi_n\rangle$$

where n_0 is the 'centre' of your wave packet and σ^2 is your energy spectrum variance expressed in terms of n. Does the wave disperse over time? Does it have a 'revival' (i.e. does it reach its original state again)?

For this section, try replacing the energy terms in the Hamiltonian as defined above with

$$E_n = k_n \frac{\hbar^2}{2m_0}, \tag{9}$$

noting the change from a quadratic dependence on k_n to linear. Does your wavepacket disperse now? Since we've changed the system Hamiltonian, we would expect the eigenstates to have changed, and indeed they are different now, but we will ignore this and continue using the eigenstates as defined in Equation 5.